

PCB Manufacturing & Assembly Capabilities

Know JLCPCB's Capabilities & Get your PCBs Built Fast

- Green
- Purple
- Red
- Yellow
- Blue
- White
- Black

Rigid PCB

Flex PCB

PCB Assembly

SMT Stencil

PCB Specifications			^
Features	Capability	Description	
Layer count	1-32 Layers	The number of	
Controlled Impedance	4/6/8/10/12/14/16/18/20/.../32 layers	User Guide to JLCPCB Impedance	
Impedance Tolerance	±10%		
	FR-4	Grade A laminates: Ya, KB, Sheng	

Material	Aluminum-Core	1-layer Alumin
	Copper-Core	1-layer copper contacts to co
	RF PCB	1 oz copper, 2- PTFE cores
FR-4 Dielectric Constants	4.5 (2-Layer PCB)	7628 Prepreg 3313 Perpreg 2116 Perpreg 4
Maximum Dimensions	FR4(1-layer): 606 × 510 mm FR4(2-layer): 670 × 600 mm FR4(4-layer): 663 × 593 mm FR4(6-layer and above): 656 × 586 mm Rogers / PTFE Teflon PCB: 590 × 438 mm Aluminum PCB: 602 × 506 mm Copper PCB: 480 × 286 mm	These limits ap mm. The thinn maximum. 2-layer FR4 PC 1020 × 600 mi 4-layer FR4 PC 1016 × 596 mn
Minimum Dimensions	FR4/Rogers/PTFE: 3 × 3 mm Castellated / Plated Edges: 10 × 10 mm Aluminum/Copper PCB: 5 × 5 mm	These limits ap mm. Manual re Panelization is boards.
Dimension Tolerance	±0.1mm	±0.1mm(Precis routing, and ±1
Thickness	0.4 – 4.5 mm	Thickness for mm (2.5 mm a only)
Thickness Tolerance	± 10%	e.g. For the 1.6 board thicknes

(Thickness≥1.0mm)		to 1.76mm(T+1
Thickness Tolerance (Thickness<1.0mm)	± 0.1mm	e.g. For the 0.8 board thicknes 0.9mm(T+0.1).
Finished Outer Layer Copper	2-layer: 1 oz / 2 oz / 2.5 oz / 3.5 oz / 4.5oz Multi-layer: 1 oz / 2 oz	
Finished Inner Layer Copper	0.5 oz / 1 oz / 2 oz	Finished copp default.
Soldermask	Green, Purple, Red, Yellow, Blue, White, and Black.	We use LPI (Li This is the mo Heat-cured int low-cost, sing
Surface Finish	HASL (leaded / lead-free), ENIG, OSP (copper core boards only)	FR4 has all thr RF boards only Aluminium cor core boards or

Drilling



Features	Capability	Description
Drill Diameter	1-layer: 0.3 – 6.3 mm 2-layer: 0.15 – 6.3 mm Multilayer: 0.15 – 6.3 mm	Holes with dia from a smaller Min. drill diam 0.15 mm (more Min. drill diam mm Min. drill diam



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Hole size Tolerance	Through-holes: +0.13 / -0.08 mm Press-fit holes: ±0.05 mm (Finished hole size: 0.55-1.025mm. Multilayer ENIG boards only. Mention the specific holes in PCB Remark)	e.g. for the 0.6 between 0.52r To avoid solde recommended
Average Hole Plating Thickness	18µm	
Hole Position Tolerance	±0.05 mm	
Blind/Buried Vias	Not supported	Currently we c make through
Min. Via hole size/diameter	0.15mm / 0.25mm	1-layer (NPTH diameter 2-layer: 0.15mm Multilayer: 0.15mm diameter ① Via diameter preferred) large ② Preferred 1 ③ 0.15mm hole and 0.2mm or less than 0.45mm corresponding
Min. Non-plated holes	0.50mm	Please draw N keep out layer.
Min. Plated Slots Width	2-layer: 0.5mm Multi-layer: 0.35mm	The minimum drawn with a p The length of 1 the width.
Min. Non-Plated Slots	1.0mm	The minimum please draw th layer(GM1 or C
	Plated: +0.13 / -0.08 mm	Plated slot is n

Slot Hole size Tolerance	Non-plated: ±0.2mm	Non-plated slot
Via Hole-to-Hole Spacing	0.2mm	
Pad Hole-to-Hole Spacing	0.45mm	
Min. Castellated Holes	0.5mm	Castellated holes, common edges, common soldered onto ① Hole diameter ② Hole to board ③ Hole to hole ④ Min. PCB size ⑤ Min. PCB thickness
Plated Edges	10 × 10mm	Plated edges and HASL is not suitable ① Min. PCB size ② Min. PCB thickness ③ At least 3 locations for edge plating and connections
		① Blind slot via

Blind Slot		② Blind slot d ③ Blind slot a width of PTH k ④ Safety dist from NPTH bli ⑤ Blind slot r (The distance the nearest inr ⑥ Supports 2 thickness of ≥
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Backdrill		Backdrill uses control the dep copper from th reducing its in ①Supports 4 thickness of ≥ ②Through-H filled after bac ③Backdrill Di than through-t ④Backdrill De customizable ⑤Dielectric T of backdrill to ⑥Safety Dist: pad/trace/cop
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Rectangular Holes / Slots	Not supported	Rectangular ho corners are nc
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Traces		^
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Features	Capability	Description
Min. track width and spacing (1 oz)	0.10 / 0.10 mm (4 / 4 mil)	1- and 2-layer: Multilayer: 0.0 acceptable in

Min. track width and spacing (2 oz)	0.16 / 0.16 mm (6.5 / 6.5 mil)	2-layer: 0.16 / Multilayer: 0.15
Min. track width and spacing (2.5 oz)	2 layer: 0.2/0.2mm(8mil/8mil)	
Min. track width and spacing (3.5 oz)	2 layer: 0.25/0.25mm(10mil/10mil)	
Min. track width and spacing (4.5 oz)	2 layer: 0.3/0.3mm(12mil/12mil)	
Track width tolerance	±20%	e.g. For a 0.1 n ranges from 0.
PTH annular ring	≧0.20mm	2-layer: 1 oz: Recomm minimum 0.18 2 oz: 0.254 m Multilayer: 1 oz: Recomm minimum 0.15 2 oz: 0.254 m
NPTH pad annular ring	≧0.45mm	Recommender 0.2 mm ring of hole for the se smaller than th the annular rin missing.
BGA	0.2mm	① 0.2mm-0.2 ENIG ② BGA pad to 0.09 mm for m ③ Vias can be filled and plate
		Minimum trace when traces a

Trace coils	0.15/0.15mm	Minimum trace when traces are (1oz). ENIG on HASL)
Hatched grid width and spacing	0.25 mm	
Same-net track spacing	0.25mm	
Inner layer via hole to copper clearance	0.2mm	
Inner layer PTH pad hole to copper clearance	0.3mm	
Pad to track clearance	0.1mm	Min. 0.1 mm (s

		mm locally for
SMD pad to pad clearance (different nets)	0.15mm	More details o SMD Compon Minimum SMD
Via hole to Track	0.2mm	
PTH to Track	0.28mm	0.35mm is rec
NPTH to Track	0.2mm	

Soldermask



Features	Capability	Description
Soldermask Expansion	1:1	LDI equipment size:solderma: production file Keep at least 6 soldermask op

Soldermask bridge	0.10mm	1oz: Min. pad spac blue, purple) Min. pad spac 2oz: Min. pad spac
Plugged vias	Filled with soldermask	Vias are filled finish. Click for detail ① Filled vias on either side ② Filled vias from other sol ③ Filled vias diameter
JLCPCB Via-in-Pad Process	Epoxy Filled & Capped Copper paste Filled&Capped	Vias are filled and then plate smooth finish. Click for detail ① Vias are fill paste filling fo conductivity. ② This proce above multilay ③ Compatible mm.
Solder mask dielectric constant	3.8	
Solder mask ink thickness	≥ 10μm	

Features	Capability	Description
Minimum Line Width	≥0.15mm	Characters will be unidentifiable.
Minimum text height	40 mil (1.0mm)	Characters height will be unidentifiable.
Character width to height ratio	1:6	The preferred ratio is 1:6.
Hollow-carved Character width to height ratio	1:6	The preferred ratio is 1:6.
Pad To Silkscreen	0.15mm	The Minimum Pad To Silkscreen is 0.15mm.

Outline



Features	Capability	Description
Routed	0.2mm	① Copper clearance: ≥0.2 mm ② Copper thickness: 0.035mm (1.37 mil) ③ Dimensional precision: ±0.2 mm (regardless of size) ④ Minimum copper thickness: 0.035mm (1.37 mil) precision, and minimum 1.5mm (0.060 inch) for larger dimensions. ⑤ Minimum size for plated through holes (PTH): 1.6mm (0.063 inch).

V-Cut	0.4mm	① Copper cle ≥ 0.4 mm ② Dimension ± 0.4 mm. PCB ③ Zero panel Alternatively, \ spacing and rc 1.6 or 2 mm bc ④ Min. panel panel dimensi ⑤ V-cut groo ⑥ Minimum sq 2mm(3mm is r
Mouse bites Panel	0.2mm	① Copper cle edges: ≥ 0.2 m ② Dimension edges: ± 0.2 m precision) ③ Panel boar ④ Serrated e ⑤ Minimum t assembly at JI mm tooling ho 3.85 mm from ⑥ Recommer 0.5mm-0.8mm the two mouse width of break with mouse-bi
Panelization with space	2mm	The spacing b as narrow spa and V-cut.

Panel of Circular PCBs

≥20mmx20mm

The single round board size ≥20mmx20mm
Panelize with 4 boards on four board

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